

Claude Code Cheatsheet

Daily essentials for maximum productivity

v3.43.0 | September 2026

How to Install

```
npm install -g @anthropic-ai/claude-code # Verify installation
which claude && claude --version
```

Health Check: `claude doctor && claude mcp list`

Essential Commands

Command	Action
<code>/help</code>	Contextual help
<code>/powerup</code>	Interactive animated lessons teaching Claude Code features
<code>/clear</code>	Reset conversation
<code>/compact</code>	Free up context
<code>/status</code>	Session state + context usage
<code>/context</code>	Detailed token breakdown
<code>/plan</code>	Enter Plan Mode (no changes)
Approve the plan	Exit Plan Mode (<code>Shift+Tab</code> to leave without approving)
<code>/model</code>	Choose model; Enter saves, s applies once
<code>/insights</code>	Usage analytics + optimization report
<code>/simplify</code>	Cleanup-only review of changed code: reuse, simplification, efficiency
<code>/batch</code>	Large-scale refactors via 5-30 parallel worktree agents
<code>/teleport</code>	Teleport session from web
<code>/tasks</code>	Monitor background tasks

Command	Action
<code>/remote-env</code>	Configure cloud environment
<code>/rc</code>	Start remote control session (Pro/Max/Team/Enterprise)
<code>/mobile</code>	Get Claude mobile app download links
<code>/fast</code>	Toggle fast mode (Opus 5.5: \$8/\$40 per MTok)
<code>/voice</code>	Toggle voice input (hold Space to speak)
<code>/recap</code>	Session context summary on return to a break
<code>/effort [level]</code>	Thinking depth: low/medium/high/xhigh/max
<code>/loop [interval] [prompt]</code>	Run a prompt on repeat (self-paced when no interval is given)
<code>/goal [condition]</code>	Autonomous multi-turn mode until condition is met
<code>/rename [name]</code>	Name or rename the current session
<code>/debug</code>	Systematic troubleshooting
<code>/exit</code>	Quit (or Ctrl+D)

Keyboard Shortcuts

Shortcut	Action
<code>Shift+Tab</code>	Cycle permission modes
<code>Esc × 2</code>	Rewind (undo)
<code>Ctrl+C</code>	Interrupt
<code>Ctrl+R</code>	Search command history

Shortcut	Action
Ctrl+L	Clear screen (keeps context)
Ctrl+B	Background tasks
Ctrl+X Ctrl+K	Kill all background agents
Alt+T	Toggle thinking where supported
Space (hold)	Voice input (requires <code>/voice</code>)
Tab	Autocomplete
Shift+Enter	New line
Ctrl+D	Exit

IDE Shortcuts: VS Code `Alt+K` | JetBrains `Cmd+Option+K`

File References

<code>@path/to/file.ts</code>	→ Reference a file
<code>@agent-name</code>	→ Call an agent
<code>!shell-command</code>	→ Run shell command

Lesser-Known Features (But Official!)

Feature	What It Does
Tasks API	Persistent task lists with dependencies
Background Agents	Sub-agents work while you code
Agent Teams	Multi-agent coordination (TeamCreate/SendMessage)
Auto-Memories	Automatic cross-session context capture
Session Forking	Rewind + create parallel timeline
LSP Tool	Code intelligence (go-to-def, refs), ~50ms vs 45s with grep
Voice Mode	Native voice input, free transcription
Remote Control	Control local session from phone/browser (Pro/Max/Team/Enterprise)
Cloud/Desktop Scheduled Tasks	Machine-off scheduling via <code>/schedule</code> or Desktop app
Skill Evals	A/B testing and trigger tuning for custom skills
Output Styles	<code>/output-style</code> : Default, Proactive, Concise, Explanatory, Learning

Feature	What It Does
Rules Templates	Ready-to-use <code>.claude/rules/</code> templates (arch/quality/test/perf)

Pro tip: These aren't "secrets", they're in the [CHANGELOG](#). Read it!

Permission Modes

Mode	Editing	Execution
Manual (<code>default</code>)	Asks	Asks
acceptEdits	Auto	Asks
Plan Mode	None	Read-only tools
auto	Classifier decides	Classifier decides
dontAsk	Only if in allow rules	Only if in allow rules
bypassPermissions	Auto	Auto (CI/CD only)

Shift+Tab to switch modes

Memory & Settings (2 levels)

Level	Path	Scope	Git
Project	<code>.claude/</code>	Team	Yes
Personal	<code>~/.claude/</code>	You (all projects)	No

Priority: Project overrides Personal

`.claude/` Folder Structure

<code>.claude/</code>	├─ CLAUDE.md	# Local memory (gitignored)
	├─ settings.json	# Hooks (committed)
	├─ settings.local.json	# Permissions (not committed)
	├─ agents/	# Custom agents
	├─ hooks/	# Event scripts
	├─ rules/	# Auto-loaded rules
	├─ architecture-review.md	├─ code-quality-review.md
	├─ test-review.md	├─ performance-review.md
	└─ skills/	# Slash commands + knowledge modules (unified)

Context Management (CRITICAL)

Statusline: `Model: Sonnet | Ctx: 89.5k | Ctx(u): 56.0%`

✅ 0-50%	⚠️ 50-70%	💡 70-90%	🔥 90%+
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Watch `Ctx(u):` → `>70%` = `/compact` , `>85%` = `/clear`

Sign	Action
Short responses	<code>/compact</code>
Frequent forgetting	<code>/clear</code>
>70% context	<code>/compact</code>
Task complete	<code>/clear</code>

Context Recovery Commands

Command	Usage
<code>/compact</code>	Summarize and free context
<code>/clear</code>	Fresh start
<code>/rewind</code>	Undo recent changes
<code>claude -c</code>	Resume last session
<code>claude -r <id></code>	Resume specific session

Plan Mode & Thinking

Feature	Activation	Usage
Plan Mode	<code>Shift+Tab × 2</code> or <code>/plan</code>	Explore without modifying
OpusPlan	<code>/model opusplan</code>	Opus for planning, Sonnet for execution

Thinking and Effort

Opus 5.5, Sonnet 5, and Fable 5.1 use adaptive thinking. Haiku 4.5 supports extended thinking but has **no effort parameter**.

Current adaptive models support `low`, `medium`, `high`, `xhigh`, and `max`. Opus 4.6 and Sonnet 4.6 support `max` but not `xhigh`.

`/effort` saves levels through `xhigh` per model. Use `s` for a session-only choice. `max` is session-only; `/effort auto` clears the active model's saved choice.

`Option+T` / `Alt+T` toggles thinking where supported. It has no effect on Opus 5.5 or Fable, whose thinking is always enabled. Lower effort for bounded work instead.

`ultrathink` adds a reasoning instruction without changing API effort. `think hard` is ordinary prompt text.

Typical Workflow

1. Start session → `claude`

2. Check context → `/status`

3. Plan Mode → `Shift+Tab × 2` (for complex tasks)

4. Describe task → Clear, specific prompt

5. Review changes → Always read the diff!

6. Accept/Reject → `y/n`

7. Verify → Run tests

8. Commit → When task complete

9. `/compact` → When context >70%

Quick Prompting Formula

WHAT: [Concrete deliverable]	WHERE: [File paths]
HOW: [Constraints, approach]	VERIFY: [Success criteria]

Example:

Add input validation to the login form.

WHERE: `src/components/LoginForm.tsx`

HOW: Use Zod schema, show inline errors

VERIFY: Empty email shows error, invalid format shows error

MCP Servers

Server	Purpose
Serena	Indexing + session memory + symbol search
grepai	Semantic search + call graph analysis
Context7	Library documentation
Sequential	Structured reasoning
Playwright	Browser automation
Postgres	Database queries
doobidoo	Semantic memory + Knowledge Graph

Serena memory: `write_memory()` / `read_memory()` / `list_memories()`

Check status: `/mcp`

Search Tools Quick Reference

Quick decision: exact text → `rg` | exact name → `rg` / Serena | concept → `grepai` | structure → `ast-grep`

Task	Tool
Find exact text	<code>rg "TODO"</code>
Find auth-related code	<code>grepai search "authentication"</code>
Who calls a function	<code>grepai trace callers "login"</code>
Get file structure	<code>serena get_symbols_overview</code>

CLI Flags Quick Reference

Flag	Usage
<code>-p "query"</code>	Non-interactive mode (CI/CD)
<code>-c / --continue</code>	Continue last session
<code>-r / --resume <id></code>	Resume specific session
<code>--teleport</code>	Teleport session from web
<code>--model sonnet</code>	Change model
<code>--add-dir ../lib</code>	Allow access outside CWD
<code>--permission-mode plan</code>	Plan mode
<code>--worktree / -w</code>	Run in isolated git worktree
<code>--max-budget-usd 5.00</code>	Max API spend limit (print mode)
<code>--dangerously-skip-permissions</code>	Auto-accept (use carefully)
<code>--debug='mcp'</code>	Debug MCP servers
<code>--allowedTools "Edit,Read"</code>	Pre-approve tools; <code>-tools</code> restricts availability
<code>--debug</code>	Debug output

Anti-patterns

❌ Don't

- Vague prompts
- Accept without reading
- Ignore warnings
- Skip permissions
- Negative constraints only

✅ Do

- Specify file + line with `@references`
- Read every diff
- Use `/compact` at 70%
- Never in production
- Provide alternatives

Cost Optimization

Current Models

Standard API rates, verified September 24, 2026. USD per million input/output tokens, excluding caching.

Model	Input / output	Context	Default effort
Haiku 4.5	\$1 / \$5	200K	Unsupported
Sonnet 5	\$2 / \$10	1M	high
Opus 5.5	\$4 / \$20	1M	medium

Model	Input / output	Context	Default effort
Fable 5.1	\$10 / \$50	1M	high

Sonnet’s published rate remains \$2/\$10 after the launch promotion. Opus 5.5 is the direct-service default; account policy can override it. Model choice does not establish a typical task cost.

Model	Use For	Cost
Haiku	Simple fixes, reviews	\$
Sonnet	Most development	\$\$
Opus	Architecture, complex bugs	\$\$\$
OpusPlan	Plan (Opus) + Execute (Sonnet)	\$\$

Quick Decision Tree

Four-Layer Quick Map

Layer	Selection question	Reference
Model	Which reasoning and cost fit the job?	Glossary
Runtime harness	Who runs the tool-observation-decision loop?	Agent Harness Map
Repository harness	Which instructions, verification, and state bound the repository?	Agent Harness Engineering
Orchestrator	Do several runtimes or sessions need coordination?	Agent Tools

Evaluate the exact model-harness pair for a coding task. Add an orchestrator when coordination is the identified need. A harness optimizer sits outside the four operating layers and changes candidate harnesses under a separate evaluation protocol.

Loop and graph are not extra substitutes. Loop names the repeated feedback and stopping rule. Graph names the executable nodes, branches, and checkpoints inside the chosen structure. For observed mechanisms, `confirmed` = visible mechanism, `claimed` = source assertion, `unknown` = point not established yet. See also [Loop & Graph Engineering](#).

Simple task	→ Just ask Claude
Complex task	→ Tasks API to plan first
Risky change	→ Plan Mode first
Repeating task	→ Create agent or command
Context full	→ <code>/compact</code> or <code>/clear</code>
Need docs	→ Use Context7 MCP
Deep analysis	→ Use Opus (thinking on by default)

The Golden Rules

- 1 Always review diffs before accepting
- 2 Use `/compact` before context gets critical (>70%)
- 3 Be specific in requests (WHAT, WHERE, HOW, VERIFY)
- 4 Plan Mode first for complex/risky tasks
- 5 Create CLAUDE.md for every project
- 6 Commit frequently after each completed task
- 7 Know what's sent: prompts, files, MCP results → Anthropic

Common Issues Quick Fix

Problem	Solution
"Command not found"	Check PATH, reinstall npm global
Context too high (>70%)	<code>/compact</code> immediately
Slow responses	<code>/compact</code> or <code>/clear</code>
MCP not working	<code>claude mcp list</code> , check config
Permission denied	Check <code>settings.local.json</code>
Hook blocking	Check hook exit code, review logic

Health Check: `which claude && claude doctor && claude mcp list`

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